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Potential ethical issues involving robot educators in South Africa

Due to several ethical and practical concerns, I would advise the south African government not introduce robot teachers at this time. While there are many areas to consider when discussing robot teachers in South Africa and from a global standpoint, I find that a significant area of concern involves the delicate balance of trust and privacy issues in the school environment, as this involves minors and many intricate relationships in the school environment which must be protected to maintain the wellbeing of the learners as well as a prosperous learning environment.

It is a main responsibility of educators all over the world to ensure a relationship of trust with their students, this involves navigating a fine line between sharing crucial information with stakeholders such as parents, guardians, other educators, and councilors while simultaneously protecting the confidentiality of the learners in question and thus protecting the relationship of trust between the teacher and the student. While certain aspects of this balance may be more easily navigated by a robot such as staying current with local and countrywide policies involving student confidentiality (including those whereas student poses a risk to themselves or others or in such instances where the educator believes that the student maybe in an abusive home environment). Other circumstances present more complex issues than can currently be solved with even a complex artificial intelligence algorithm, for example “what is ethical behavior under professional codes of ethics is not necessarily legal and vice versa”. (1) These rare but plausible instances are far more likely to occur in a school environment and could lead to an infinite loop in processing and subsequent indecision, or an incorrect decision could have catastrophic consequences for the teacher-learner relationship or the parent-teacher relationship.

As in the story Melokuhle where the care Bot was programmed to follow the users ethical view, and use this to impose rules, when considering robots (care bots) they are generally considered to be programmed to follow an ethical principle (either pre-defined by the programmers or set by a

user), as well as follow the laws of the relevant country. As in the previous quote, issues with regards to minors may cause circumstances to arise which have no clear solution and which may present a conflict between the law and the ethical principle programmed. These situations are distinctly human, and I am unaware of any current technological advancements which allow robots to think in a manner which would allow for navigation of such complexly human challenges. This being mainly due to the fact that one cannot program human experience and a human manner of thinking and processing.

Furthermore, privacy issues arise when sharing information which may be considered private to the student, with any relevant stakeholders. Educators, much like councilors, depend on a relationship of trust with their learners to provide quality education, however many parents believe that their children do not have a right to privacy when it comes to their education. This issue is one which educators spend much time navigating (3), and its solution is unique to every learner-parent relationship. A robot teacher would need to be programmed in such a way as to determine what information is crucial for parents and other stakeholders to be made aware of and what information should be kept confidential. This extends to conversations which may not have been directly between the robot and the learner but includes conversations which may have been overheard by the robot, even without the learner's knowledge.

This further leads to issues involving legal action, as it is currently unclear what information a robot may store, for how long that information should be stored and in what circumstances law enforcement or any other party would have a right to access this private information. (2) Further legal issues involve the storage of learner's information, as robots create increased risk of hacking or any other privacy leaks which may occur with any electronically stored personal information. (2) while this may not be a direct risk, a student who is aware of this risk however minimal maybe reluctant to share certain private information due to the fact that others could become aware by direct or indirect methods through the robot teacher.

And finally, robot teaches would need to be aware of any people within earshot when speaking with a student about confidential information. (2) being a school, and thus a busy environment there may be several people who are not privileged to the students private information within earshot at any given time and thus preventing effective communication between a robot teacher and its student.

While most of these issues are universal and not localized to South Africa, they should be very seriously considered when deliberating the implementation of robot teachers in South Africa and worldwide. Due to these issues as well as others, I do not believe that it would be wise to introduce robot teachers in South Africa until such a time as these issues can be effectively handled with the use of AI technology in other settings, and those solutions clearly present a smooth transition into the teaching environment, as we cannot risk the education of our youth.

(1) Sealander, K.A., Schwiebert, V.L., Oren, T.A. and Weekley, J.L., 1999. Confidentiality and the law. *Professional School Counseling*, *3*(2), pp.122-127.

(2) Jokinen, K. and Wilcock, G., 2021, August. Do you remember me? ethical issues in long-term social robot interactions. In *2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)* (pp. 678-683). IEEE.

(3) Dhirani, L.L., Mukhtiar, N., Chowdhry, B.S. and Newe, T., 2023. Ethical dilemmas and privacy issues in emerging technologies, *23*(3), p.1151.